

Differences Between Temp Table vs Table Variable

1. Definition and Scope

Temporary Tables: Created using CREATE TABLE with names prefixed by # (local) or ## (global). Stored in tempdb.

Table Variables: Declared using DECLARE @TableName TABLE (...). Scope is limited to the current batch, function, or stored procedure.

2. Lifetime

Temporary Tables: Exist until dropped explicitly or the session ends.

Table Variables: Automatically removed when the batch, function, or stored procedure finishes.

3. Performance and Usage

Temporary Tables: Better for large datasets, joins, indexes, and statistics.

Table Variables: Better for small datasets with lower overhead but fewer optimizer features.

4. Transaction Logs

Temporary Tables: Fully logged and participate in transactions.

Table Variables: Minimally logged and changes are not rolled back when the surrounding transaction is rolled back.

5. Use Cases

Temporary Tables: Best for complex queries, large data, and rollback support.

Table Variables: Best for small, temporary datasets and simple processing.